

## Single Organ Segmentation (Model=SegFormer)

(Framework=PyTorch, Encoder=resnet50, Threshold=0.40, trainig size=0.8, test\_size=0.1, vaidation size=0.1)

a) **Intestinal Veins Data** (Train image=853, Test image =106, validation image=108)

image size 128x128 (Model: SegFormer)

| Number of Epochs       | 10 Epoch | 30 Epoch | 50 Epoch |
|------------------------|----------|----------|----------|
| IoU                    | 0.4962   | 0.5201   | 0.5692   |
| Precision              | 0.6039   | 0.6092   | 0.6704   |
| Recall                 | 0.7333   | 0.7715   | 0.7546   |
| Specificity            | 0.9949   | 0.9957   | 0.9971   |
| F1 Skore               | 0.6345   | 0.6612   | 0.6945   |
| Running Time (seconds) | 375.44   | 1166.39  | 1770.61  |

image size 256x256 (Model: SegFormer)

| Number of Epochs       | 10 Epoch | 30 Epoch | 50 Epoch |
|------------------------|----------|----------|----------|
| IoU                    | 0.5265   | 0.5597   | 0.6500   |
| Precision              | 0.5696   | 0.6055   | 0.7162   |
| Recall                 | 0.8620   | 0.8692   | 0.8684   |
| Specificity            | 0.9939   | 0.9944   | 0.9965   |
| F1 Skore               | 0.6710   | 0.6971   | 0.7736   |
| Running Time (seconds) | 407.36   | 1232.22  | 1771.55  |

b) **Ureter Data** (Train image=1020, Test image=128, validation image=127)

image size 128x128 (Model: SegFormer)

| Number of Epochs       | 10     | 30     | 50      |
|------------------------|--------|--------|---------|
| IoU                    | 0.4280 | 0.4720 | 0.5485  |
| Precision              | 0.6180 | 0.6963 | 0.6123  |
| Recall                 | 0.5650 | 0.5856 | 0.8048  |
| Specificity            | 0.9962 | 0.9974 | 0.9950  |
| F1 Skore               | 0.5467 | 0.5916 | 0.6782  |
| Running Time (seconds) | 392.82 | 973.48 | 2046.38 |

image size 256x256 (Model: SegFormer)

| Number of Epochs       | 10     | 30      | 50      |
|------------------------|--------|---------|---------|
| IoU                    | 0.4630 | 0.5455  | 0.5704  |
| Precision              | 0.5214 | 0.7664  | 0.6368  |
| Recall                 | 0.7868 | 0.6513  | 0.7946  |
| Specificity            | 0.9926 | 0.9981  | 0.9959  |
| F1 Skore               | 0.5973 | 0.6676  | 0.6926  |
| Running Time (seconds) | 427.35 | 1168.73 | 2012.62 |

c) **Inferior mesenteric artery data** (Train image=872, Test image=109, validation image=109)

image size 128x128 (Model: SegFormer)

| Number of Epochs       | 10            | 30            | 50             |
|------------------------|---------------|---------------|----------------|
| IoU                    | 0.5321        | 0.5992        | 0.6207         |
| Precision              | 0.5843        | 0.7388        | 0.7151         |
| Recall                 | 0.8460        | 0.7541        | 0.7960         |
| Specificity            | 0.9859        | 0.9947        | 0.9932         |
| F1 Skore               | 0.6737        | 0.7229        | 0.7425         |
| Running Time (seconds) | <b>379.47</b> | <b>863.67</b> | <b>1823.74</b> |

image size 256x256 (Model: SegFormer)

| Number of Epochs       | 10            | 30             | 50             |
|------------------------|---------------|----------------|----------------|
| IoU                    | 0.5552        | 0.6201         | 0.6410         |
| Precision              | 0.6287        | 0.7545         | 0.7264         |
| Recall                 | 0.8382        | 0.7397         | 0.8378         |
| Specificity            | 0.9882        | 0.9953         | 0.9926         |
| F1 Skore               | 0.7005        | 0.7342         | 0.7605         |
| Running Time (seconds) | <b>408.85</b> | <b>1098.11</b> | <b>1825.51</b> |

d) **Vglands Data** (Train image=**958**, Test image=**121**, validation image=**119**)

image size 128x128 (Model: SegFormer)

| Number of Epochs       | 10            | 30             | 50             |
|------------------------|---------------|----------------|----------------|
| IoU                    | 0.4117        | 0.4621         | 0.5392         |
| Precision              | 0.6396        | 0.6304         | 0.6576         |
| Recall                 | 0.5241        | 0.5775         | 0.7061         |
| Specificity            | 0.9933        | 0.9948         | 0.9920         |
| F1 Skore               | 0.5259        | 0.5740         | 0.6472         |
| Running Time (seconds) | <b>375.13</b> | <b>1206.53</b> | <b>1912.36</b> |

image size 256x256 (Model: SegFormer)

| Number of Epochs       | 10            | 30             | 50             |
|------------------------|---------------|----------------|----------------|
| IoU                    | 0.5121        | 0.5557         | 0.5949         |
| Precision              | 0.6665        | 0.6588         | 0.7171         |
| Recall                 | 0.6275        | 0.7392         | 0.7565         |
| Specificity            | 0.9949        | 0.9936         | 0.9951         |
| F1 Skore               | 0.6194        | 0.6746         | 0.7167         |
| Running Time (seconds) | <b>408.73</b> | <b>1354.31</b> | <b>1973.80</b> |

e) **Pancreas Data** (Train image=**938**, Test image=**117**, validation image=**118**)

**image size 128x128 (Model: SegFormer)**

| Number of Epochs       | 10            | 30            | 50 (46 e.s.)   |
|------------------------|---------------|---------------|----------------|
| IoU                    | 0.4942        | 0.5122        | 0.5797         |
| Precision              | 0.6805        | 0.6922        | 0.6828         |
| Recall                 | 0.6200        | 0.6293        | 0.7783         |
| Specificity            | 0.9941        | 0.9956        | 0.9938         |
| F1 Skore               | 0.6104        | 0.6293        | 0.7035         |
| Running Time (seconds) | <b>370.31</b> | <b>897.63</b> | <b>1736.12</b> |

**image size 256x256 (Model: SegFormer)**

| Number of Epochs       | 10            | 30            | 50             |
|------------------------|---------------|---------------|----------------|
| IoU                    | 0.5278        | 0.5494        | 0.5901         |
| Precision              | 0.6193        | 0.7118        | 0.7297         |
| Recall                 | 0.7317        | 0.6824        | 0.7451         |
| Specificity            | 0.9903        | 0.9954        | 0.9937         |
| F1 Skore               | 0.6504        | 0.6723        | 0.7140         |
| Running Time (seconds) | <b>403.63</b> | <b>948.24</b> | <b>1958.42</b> |

**image size 512x512 (Model: SegFormer)**

| Number of Epochs       | 10             | 30             | 50             |
|------------------------|----------------|----------------|----------------|
| IoU                    | 0.5516         | 0.6398         | 0.6831         |
| Precision              | 0.6508         | 0.7183         | 0.7725         |
| Recall                 | 0.7749         | 0.8104         | 0.8559         |
| Specificity            | 0.9895         | 0.9940         | 0.9942         |
| F1 Skore               | 0.6813         | 0.7473         | 0.7997         |
| Running Time (seconds) | <b>1080.54</b> | <b>3106.61</b> | <b>5849.35</b> |